

# GEORGI KRASTEV

Senior Full-Stack Engineer | Technical Architect | Platform Modernization Specialist

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## PROFESSIONAL SUMMARY

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**Hands-on technical architect** with 9+ years architecting and shipping scalable, high-impact platforms across **real-time systems** (satellites), **industrial automation** (robots), **fintech integrations** (payments), and **AI-driven automation**. Proven expertise in **legacy platform modernization**, **design system leadership**, **third-party integrations at scale**, and **rapid MVP delivery** (3–4x faster than industry standard).

### Core competencies:

- Full-stack development (React/Vue/Node/AWS)
- System architecture & technical decision-making
- Design systems & component standardization
- Payment processing & SOAP/REST integrations
- Real-time data visualization & WebSocket systems
- Cloud infrastructure (AWS, GitHub Actions CI/CD)
- AI/automation workflows (n8n, custom nodes, LLMs)
- Cross-functional team leadership & vendor management

**Known for:** Shipping MVPs faster than competitors, solving complex integration problems, centralizing fragmented codebases, and driving measurable operational improvements (cost reduction, time savings, quality gains).

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## CORE TECHNICAL SKILLS

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### Frontend Architecture

**React** (Hooks, Context, advanced patterns), **Redux** (state management at scale), **TypeScript** (strict typing, interfaces), **Vue 3** (Composition API, modern patterns), **Vite** (build optimization), **Pinia** (Vue state), **Tailwind CSS** (utility-first styling), **Material UI**, **Storybook** (component documentation), **WebSockets** (real-time bidirectional communication)

## Backend & Database

**Node.js** (server-side JavaScript, high concurrency), **Express.js** (HTTP framework), **MongoDB** (document database, flexible schema), **SQL** (relational databases, complex queries), **Supabase** (PostgreSQL + realtime APIs), **REST APIs** (design & implementation), **SOAP/XML** (legacy integrations)

## Cloud & DevOps

**Amazon Web Services (AWS)**: S3 (object storage), CloudFront (CDN), EC2 (compute), Cognito (identity), Lambda (serverless), RDS (managed database); **Docker** (containerization, multi-stage builds), **GitHub Actions** (CI/CD pipelines, multi-environment deployments), **AWS Authentication & User Management** (IAM, session handling)

## AI & Automation

**n8n** (workflow orchestration, 50+ step workflows), **Custom JavaScript Nodes** (extending n8n with bespoke logic), **OpenAI API** (GPT integrations, prompt engineering), **Automated Reporting Agents** (data aggregation, trend analysis, PDF generation), **IVR & Voice AI** (AI-generated voice responses)

## Architecture & Leadership

**Design Systems** (Storybook, component libraries, brand consistency), **Legacy Modernization** (rewriting monoliths, architectural refactors), **Third-Party Integrations** (payment processors, APIs, webhooks), **RBAC** (role-based access control), **OAuth 2.0** (secure authentication flows), **Reconciliation Logic** (financial data alignment), **Agile/Scrum**, **System Design**

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# PROFESSIONAL EXPERIENCE

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## PRINCIPAL FULL-STACK ENGINEER / SOLUTION ARCHITECT

**Boutique Tech Consultancy (MVP Forge / ZenGroup Umbrella)** | Sofia, Bulgaria | *January 2024 – Present*

Leading technical delivery of **AI automation services** and **rapid SaaS development** for clients, with focus on **workflow automation**, **payment integrations**, and **MVP velocity**.

**Prezaredi.bg – Fuel Payments & Fleet Cards Platform** (*Feb 2024 – Present*)

**Problem:** Corporate clients and end-users needed a digital alternative to plastic fuel cards at petrol stations—reducing friction, enabling delayed payments, and providing real-time fleet

management.

## Architecture & Scope:

- **Mobile App:** React Native (iOS/Android) for fleet card and go-card users; cross-platform native performance.
- **Backend:** Node.js/Express running on AWS EC2; **PostgreSQL** database for transactional consistency; RDS for managed persistence.
- **Payment Processor:** BORICA (Bulgaria's largest card payment provider) SOAP XML API for card issuance, printing, PIN delivery, and settlement.
- **Third-Party Integrations:** Petrol station provider APIs for fuel price feeds, transaction capture, and real-time balance updates.
- **CI/CD Infrastructure:** GitHub Actions end-to-end (dev, staging, production); automated testing, linting, and deployment gates.
- **Call Center Workflow:** Lost/stolen card blocking via IVR with **AI-generated voice responses** (natural language, no human agent required after hours).

## Key Technical Decisions & Risk Mitigation:

- **Reconciliation Logic:** Solved critical matching problems where transaction amounts differed between Prezaredi and petrol station ledgers (e.g., 55€ vs 55.50€ during invoicing). Implemented **idempotent reconciliation engine** with tolerance thresholds and manual review queues.
- **SOAP/XML Handling:** Built XML parsing and error recovery for BORICA API (strict schema validation, retry logic, webhook verification).
- **Offline Sync:** React Native app caches card balance and transaction history; syncs on reconnect with conflict resolution.
- **Security:** AWS Cognito for user authentication; encrypted card data in transit (TLS 1.3); PCI scope reduction through AWS tokenization.

## Metrics & Outcomes:

- **20 corporate clients** onboarded; **~400 active fleet cards** in circulation.
- **Thousands of transactions per month** processed; **99%+ uptime** maintained.
- **Reduced petrol station card issuance time** from 5–7 days to real-time digital activation.
- **Delayed payment + discount model** improved cash flow for corporate clients; increased adoption vs. cash-only competitors.

## Team Coordination:

- Managed **~15 external developers, designers, and project managers** from vendor organization.
- Owned **architecture decisions, integration contracts, acceptance criteria, and quality gates**.

- Negotiated with **petrol station technical teams** and **C-level brand directors** to secure API access, commercial terms, and integration roadmaps.
  - Built landing page and conducted user research to refine feature priorities.
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### Automated Quarterly Reporting Agent (n8n + AWS) *(Ongoing)*

**Problem:** Clients running survey campaigns needed monthly/quarterly trend analysis, but manual compilation took **60–80 hours per month** (equivalent to 1–2 FTEs).

#### Architecture:

- **50-step n8n workflow** orchestrating data ingestion, transformation, and reporting.
- **Custom JavaScript Nodes** (proprietary logic) for statistical analysis and trend detection.
- **Data Pipeline:** Survey platform → MongoDB → aggregation → AWS Lambda for PDF generation.
- **Storage:** PDFs saved to S3; AWS CloudFront CDN for fast downloads.
- **Distribution:** React UI for self-service downloads; automated email delivery (SES).

#### Key Components:

1. **Data Collection:** Fixed-period ingestion (monthly, quarterly); de-duplication and validation.
2. **Trend Analysis:** Compares current period against historical baselines; flags improving/declining metrics.
3. **AI-Powered Insights:** Uses OpenAI API to generate actionable suggestions (e.g., "Response rate down 12%; consider shortening survey").
4. **PDF Generation:** Dynamic report with charts, tables, and recommendations; branded with client logo.

#### Metrics & Outcomes:

- **Saves 60–80 hours per month** (2 FTEs freed for higher-value work).
  - **Reduced report delivery time** from 2–3 days to minutes.
  - **100% accuracy** (no manual compilation errors).
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### TabiSurvey – Advanced Survey Platform MVP *(Ongoing)*

**Problem:** Clients needed a modern, flexible survey builder with rich question types, RBAC, and advanced analytics—faster than building from scratch.

#### Architecture:

- **Frontend:** Vue 3 (Composition API) + Vite (fast dev experience) + Pinia (state management).

- **UI Components:** Tailwind CSS for rapid styling; Material UI for pre-built accessibility.
- **Backend:** Supabase (PostgreSQL + realtime subscriptions) for rapid API iteration.
- **Database Schema:** Surveys, questions (multiple types), responses, user roles, access control.

Features:

- **Survey Builder:** Drag-and-drop interface; support for text, multiple-choice, emoji reactions, star ratings.
- **RBAC:** Role-based access (admin, editor, viewer); team collaboration.
- **Analytics Dashboard:** Real-time response counts, sentiment analysis, comparative trends.
- **Data Export:** CSV/JSON export for integration with BI tools.

Metrics & Outcomes:

- **MVP shipped in 3 weeks** (typical build time: 8–12 weeks = 4x faster).
- Production-ready performance (Lighthouse 95+).
- Ready for immediate client onboarding and feedback iteration.

TECHNICAL LEAD & PLATFORM ARCHITECT

EnduroSat | Sofia, Bulgaria | *December 2022 – December 2025*

*(Official title: Product Manager; presented as Technical Lead & Platform Architect due to hands-on engineering contribution.)*

Led the **complete architectural overhaul** of EnduroSat's core Satellite Operations Platform, stabilizing a fragile legacy system into a **high-performance, scalable, maintainable platform** serving mission-critical operations.

Satellite Operations Platform V2 – Complete Rewrite

**Problem:** The legacy platform was unreliable, difficult to maintain, and lacked features needed for modern satellite operations. Rewrite required from scratch.

Architecture (Before → After):

| Aspect             | Before                    | After                                       |
|--------------------|---------------------------|---------------------------------------------|
| Frontend Framework | Legacy<br>jQuery/Backbone | <b>React</b> (Hooks, Context)               |
| State Management   | Scattered global state    | <b>Redux</b> (normalized schema, dev tools) |

| Aspect         | Before                   | After                                                |
|----------------|--------------------------|------------------------------------------------------|
| Styling        | Inline CSS, BEM          | <b>Tailwind CSS</b> + Material UI components         |
| Build Tool     | Webpack (slow dev)       | Vite (instant HMR)                                   |
| Performance    | Slow telemetry rendering | <b>WebSockets</b> for real-time updates; memoization |
| AWS Deployment | Manual deploys           | GitHub Actions CI/CD (dev/staging/prod)              |

Key Features Implemented:

- 1. **Live Ground Track:** Real-time visualization of satellite orbital path; WebSocket updates from server; interactive 3D globe (Cesium.js or similar).
- 2. **Command Center:** Unified interface for scheduling satellite commands; form validation; error handling; command history.
- 3. **Pass Schedule:** Display upcoming satellite passes; time-to-LOS (loss of signal) countdown; predictive analytics.
- 4. **Telemetry Dashboard:** Real-time sensor readings (temperature, power, communication status); historical trends; alerting.

Metrics & Outcomes:

- **Performance improvement:** Reduced initial load time from 8s → 2s; live telemetry refresh from 5s delay → <500ms.
- **Reliability:** Platform stabilized from frequent crashes to 99.9%+ uptime during mission-critical operations.
- **Development velocity:** New feature delivery improved 40% due to cleaner architecture.

Component Library & Design System Centralization

**Problem:** EnduroSat had multiple internal "UI Ops" products (monitoring dashboards, mission planning tools, etc.), each with its own button styles, colors, and layouts. Code duplication was rampant; bugs were fixed in one product but not others.

**Solution:** Built a **centralized Storybook component library** and enforced brand guidelines across all products.

Deliverables:

- **Storybook:** Interactive component documentation; 30+ reusable components (Button, Input, Modal, Table, Chart wrapper, etc.).

- **Brand Guidelines:** Centralized color palette, typography, spacing rules, and accessibility standards (WCAG 2.1 AA).
- **Testing:** Unit tests for all components (Jest); visual regression tests.
- **CI/CD Integration:** Design system changes trigger automated tests before merge.

### Adoption & Impact:

- **20–25% codebase reduction** across all 5 internal products (duplicate component logic centralized).
  - **Bug reduction:** Shared components tested once; fixes applied globally.
  - **Onboarding:** New developers can spin up UI in hours, not days.
  - **Maintenance burden:** 1 design system team manages components for all products.
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### Customer Success Portal (MyEnduroSat)

**Problem:** Customers had no visibility into their satellite health, service updates, or support status. Support tickets were high due to lack of self-service options.

**Solution:** Built a customer-facing portal using the new design system.

### Features:

- **Dashboard:** Satellite health overview, last contact time, upcoming passes, mission status.
- **Service Updates:** Real-time notifications of platform maintenance, new features, and bugs.
- **Support Portal:** Self-service ticket submission, status tracking, knowledge base search.
- **Billing & Usage:** Usage metrics, invoice history (if applicable).

### Metrics & Outcomes:

- **Support tickets reduced by ~20%** (self-service answered most common questions).
  - **Customer satisfaction:** NPS improved due to transparency and self-service empowerment.
  - **Operational efficiency:** Support team freed up for higher-value, complex issues.
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### AWS Infrastructure & CI/CD

#### Services Used:

- **S3:** Frontend static assets (HTML/CSS/JS bundles).
- **CloudFront:** CDN distribution; global caching; DDoS protection.
- **EC2:** Backend API servers (Node.js); auto-scaling groups.
- **Cognito:** User authentication; session management; multi-factor authentication.

- **RDS:** PostgreSQL database; automated backups; read replicas.
  - **GitHub Actions:** Build → Test → Deploy pipeline; environment-specific configurations.
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## SENIOR FRONTEND ENGINEER

**DevCloud BG** | Plovdiv, Bulgaria | 2019 – 2022

Industrial robotics company manufacturing silicon-wafer processing systems. Built the **Robot Control Center (RCC)**, a web application enabling operators to control expensive, complex machinery through an intuitive no-code interface.

### Robot Control Center (RCC) – WebSocket-Based Control System

**Problem:** Operating industrial robots required deep technical knowledge; operators needed weeks of training. Manual command input was error-prone and slow.

**Solution:** Built a visual "block puzzle" workflow editor that abstracts robot commands into draggable blocks.

#### Architecture:

- **Frontend:** React + TypeScript; WebSockets for persistent, low-latency communication.
- **Workflow Builder:** Drag-and-drop interface; block types (Move, Wait, Sense, Conditional, Loop).
- **Execution Engine:** Translates block sequences into robot bytecode; sends via WebSocket.
- **Real-Time Feedback:** Robot status updates streamed back (position, temperature, errors).

#### Key Technical Challenges Solved:

1. **Low-Latency Control:** Commands must execute within 100ms of user interaction (safety-critical). Implemented WebSocket compression and binary protocol.
2. **Offline Support:** Robot can cache queued commands during network loss; syncs on reconnect.
3. **Safety Validation:** Blocks are validated before execution (no invalid sequences, no out-of-bounds moves).

#### Metrics & Outcomes:

- **Operator training time reduced** from 4 weeks → 2 weeks.
  - **Error rate reduced** by 90% (no manual command syntax errors).
  - **Throughput increased** by ~40% (faster command execution).
  - **On-site deployment:** Traveled to China for 1 month; deployed, tested, and trained operators in live manufacturing environment.
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# SOFTWARE ENGINEER

**Aucoda** | Manchester, UK | *2017 – 2019*

Contributed to a **cross-platform application development platform** that compiles high-level source code into production-ready iOS, Android, and web applications.

## AU Programming Language – OAuth & Cross-Platform Integration

**Problem:** Generated apps needed secure authentication; implementing OAuth separately for each platform introduced inconsistencies and bugs.

**Solution:** Built a unified OAuth handler in the AU language runtime that generates identical auth logic across all platforms.

### Work:

- **OAuth 2.0 Implementation:** Authorization code flow, token refresh, secure storage of credentials.
- **Platform Bridging:** JavaScript (for web) ↔ native Objective-C (for iOS) interop; custom serialization.
- **Syntax Design:** Proposed language syntax for OAuth configuration (host, client ID, scopes, redirect URI).
- **Testing:** Unit tests across platforms; integration tests with real OAuth providers (Google, GitHub).

### Outcome:

- Generated apps shipped with OAuth support; reduced auth-related bugs by ~80%.
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# EDUCATION

## Bachelor of Science in Computer Science

University of Manchester | *2015 – 2017*

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# LANGUAGES

- **Bulgarian** – Native fluency
  - **English** – Professional fluency (written, verbal, technical)
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# KEY ACHIEVEMENTS & QUANTIFIED IMPACT

| Achievement               | Metric                                          | Impact                                 |
|---------------------------|-------------------------------------------------|----------------------------------------|
| Prezaredi.bg Launch       | 20 clients, 400 fleet cards, 99%+ uptime        | Enabled digital fuel payments at scale |
| Automated Reporting Agent | 60–80 hours/month saved                         | 2 FTEs freed for strategic work        |
| TabiSurvey MVP            | 3 weeks to production                           | 4x faster than industry standard       |
| EnduroSat Platform V2     | 8s → 2s load time; 5s → <500ms telemetry        | 75% faster; mission-critical stability |
| Design System             | 20–25% code reduction                           | Fewer bugs, faster development         |
| MyEnduroSat Portal        | 20% reduction in support tickets                | Improved customer self-service         |
| RCC Deployment            | 4 weeks → 2 weeks training; 90% error reduction | Operator productivity up 40%           |